

1. Diketahui : $y = 4x - x^2$, sumbu x ($y=0$)

Diputar mengitari sumbu y

Hitung volume benda putar

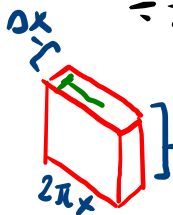
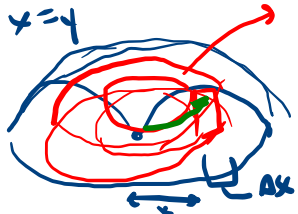
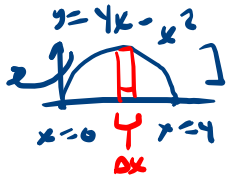
$$y = 4x - x^2 = -x^2 + 4x$$



$$-x^2 + 4x = y = 0$$

$$-x^2 + 4x = 0 \rightarrow x^2 - 4x = 0$$

$$x(x-4) = 0 \quad \begin{cases} x=0 \\ x=4 \end{cases}$$



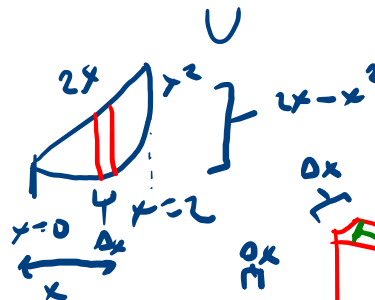
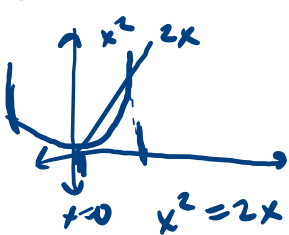
$$\Delta V = (2\pi x)(4x - x^2) \Delta x$$

$$V = 2\pi \int_0^4 (4x^2 - x^3) dx$$

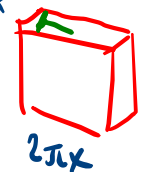
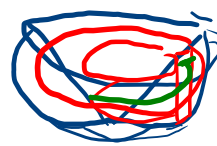
$$= 2\pi \left[4 \cdot \frac{x^3}{3} - \frac{x^4}{4} \right]_0^4$$

$$\begin{aligned} &= 2\pi \left[\left(4 \cdot \frac{4^3}{3} - \frac{4^4}{4} \right) - 0 \right] \\ &= 2\pi \left[\frac{256}{3} - \frac{256}{4} \right] = 512\pi \left(\frac{1}{3} - \frac{1}{4} \right) = 512\pi \left(\frac{4}{12} - \frac{3}{12} \right) = 512\pi \cdot \frac{1}{12} \\ &= 128\pi \end{aligned}$$

2. Bounded: $y = 2x$, $y = x^2$. Diploid symmetry



$$\Delta V = 2\pi x (2x - x^2) \Delta x$$



$$x^2 = 2x$$

$$x^2 - 2x = 0$$

$$x(x - 2) = 0$$

$$x = 0, x = 2$$

$$V = 2\pi \int_0^2 (2x^2 - x^3) dx = 2\pi \left[2 \cdot \frac{x^3}{3} - \frac{x^4}{4} \right]_0^2 = 2\pi \left(\frac{2 \cdot 2^3}{3} - \frac{2^4}{4} \right)$$

$$= 2\pi \left(\frac{16}{3} - \frac{16}{4} \right) = 32\pi \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{32\pi}{12} = \frac{8\pi}{3}$$